

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

16th March 2018

Subject SA5 – Finance

Time allowed: Three hours (14.45* - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** BBI Limited is a very large listed public sector bank in Bharatia which is one of the fastest growing and largest economies in the world. Currently the Government of Bharatia owns 70% of the equity of BBI; large institutional investors (both domestic and international) hold about 20% while the balance 10% is widely held by the public.

BBI has an unlisted Life Insurance Subsidiary. BBI owns 74% of the equity capital of this subsidiary and the balance 26% is owned by an internationally well-known life insurance company.

The economy of Bharatia is in a high interest rate regime. The Government of Bharatia has been prodding the Central Bank of the country to reduce the bench mark interest rates.

- i)** What are the implications of high interest rates to the economy? (8)

- ii)** The Central Bank has been contemplating to reduce the interest rates.

- a)** What actions the Government is expected to take to influence the Central Bank to reduce the interest rates? (5)

- b)** How does lowering of interest rates help the economy and the Bank's Balance Sheet? (6)

The Central Bank of the country is currently in the process of transitioning from the Basel II framework to the Basel III framework. It has encouraged all the banks in the country to go in for an early adoption of Basel III framework. BBI has opted for early adoption of Basel III

- iii)** Briefly describe the key additional requirements under Basel III compared to Basel II; and the plausible reasons for BBI to opt for early adoption of Basel III (8)

BBI is currently having a credit rating of AA. It is looking at improving its rating to AAA.

- iv)** What are the key measures BBI needs to take to improve its rating; and what are the advantages and disadvantages of the improved rating from the stand point of the bank? (7)

As part of the exercise for improving its credit rating, BBI is looking at augmenting its Tier 1 capital position. The Bank is contemplating to realize value from its life insurance subsidiary as one of the options for augmenting the Tier 1 capital.

- v)** Outline the alternatives available to the bank for realizing value from the subsidiary (3)

BBI has divested 30% of its stake in the life insurance subsidiary through a public issue of shares at an issue price of Rs.800 per share [Face value per share is Rs.10]. A year after listing, the shares of this listed life insurance subsidiary is being quoted at Rs.650 per share

- vi)** Briefly outline the key factors that might have led to the fall in share price; and what measures the life insurance subsidiary must take to improve the share price (7)

BBI is now planning to venture into the general insurance [GI] market using disruptive technology. For this purpose, BBI intends to set up a separate General Insurance Subsidiary with 100% of the capital held by the Bank and few private equity investors. During the initial phase this GI subsidiary will focus on only two lines of business – health and motor. The only

distribution channel that will be used for selling these products is a mobile based application called BBIAPP.

- vii)** You have been engaged as consultant to evaluate this business initiative. Briefly discuss the sources of upside potential and the downside risks of the proposed initiative (8)
 - viii)** The CFO of the bank has convened a meeting with you to discuss the possibility of putting in place a robust appraisal framework for evaluating other business opportunities in the GI space using disruptive technology as the primary enabler. Discuss the key considerations that need to be taken into account for appraising such business opportunities (7)
 - ix)** The General Insurance Subsidiary proposes to get into a 50% quota share of reinsurance arrangement with the country's largest reinsurer for the next three years. Discuss the benefits and drawbacks of such reinsurance arrangement from the stand points of the insurer and the reinsurer (6)
- [65]**

Q. 2) A global pharmaceutical company with operations in India has recently set up a separate medical technology company in India.

The new company will focus exclusively on developing products that deliver cheaper, faster and more efficient patient care. Currently this company is exploring the possibility of manufacturing products based on cutting-edge emerging technologies in the Bio- Engineering space.

The company has engaged a consultant to evaluate the financial feasibility of launching these products. The consultant has used a deterministic financial evaluation model with scenario testing. The financial evaluation done by the consultant reveals that the proposed project has a negative net present value even under the most optimistic scenario

The senior leadership team of the company is currently discussing the findings of the feasibility study.

- i)** The R&D (Research and Development) Director of the company states that financial appraisal techniques cannot be used for assessing inventions and innovations. Evaluate this statement (5)
- ii)** The CFO of the company states that the conventional NPV approach cannot be used for this project because this project has embedded "real options"
 - a)** Define the term "real option" and explain how a real option differs from a financial option (3)
 - b)** Briefly discuss the plausible real options associated with this project. (5)
 - c)** What are the drawbacks of applying the "Net Present Value" approach for assessing projects with real options (2)
 - d)** The CFO states that the real options associated with this project needs to be valued using the Black and Scholes Option Pricing Model (BSOPM). Evaluate this statement (5)

The leadership team decides to disregard the consultant's report and decides to undertake a thorough in-house evaluation of the risks and rewards associated with this project

The CFO of the company states that the project's financial feasibility must be assessed using a stochastic model rather than using a deterministic model

- iii)** Evaluate the relative merits and demerits of using a stochastic model versus a deterministic model; and recommend an appropriate evaluation model for the given project. (7)

The CFO of the company also informs the leadership team that, given the nature of the project, it will be difficult to raise equity and/or debt at an acceptable cost for financing the project. Hence, he proposes that the bulk of the funding requirement for the project needs to be met through a mix of domestic and cross-border financial lease arrangements

- iv)** Evaluate the merits and demerits of this proposal (8)
[35]
